# P-Channel 30 V (D-S) 175 °C MOSFET

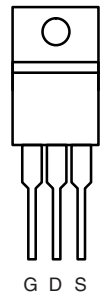
| PRODUCT SUMMARY | | |
|---|---|---|
| $V_{DS}$ (V) | $R_{DS(on)}$ (Ω) | $I_D$ (A)[a] |
| - 30 | 0.007 at $V_{GS}$ = - 10 V | ± 75 |
| | 0.010 at $V_{GS}$ = - 4.5 V | ± 75 |

## FEATURES

- Compliant to RoHS Directive 2002/95/EC

Available
RoHS* COMPLIANT

**TO-220AB**

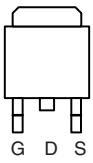



Top View

SUP75P03-07

DRAIN connected to TAB

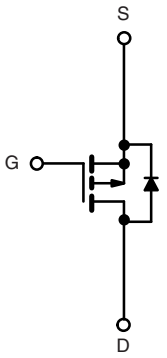
**TO-263**




Top View

SUB75P03-07

**Ordering Information:** SUB75P03-07 (TO-263)
SUB75P03-07-E3 (TO-263, Lead (Pb)-free)
SUP75P03-07 (TO-220AB)
SUP75P03-07-E3 (TO-220AB, Lead (Pb)-free)




P-Channel MOSFET

| ABSOLUTE MAXIMUM RATINGS ($T_C$ = 25 °C, unless otherwise noted) | | | | |
|---|---|---|---|---|
| **Parameter** | | **Symbol** | **Limit** | **Unit** |
| Gate-Source Voltage | | $V_{GS}$ | ± 20 | V |
| Continuous Drain Current ($T_J$ = 175 °C) | $T_C$ = 25 °C | $I_D$ | - 75[a] | A |
| | $T_C$ = 125 °C | | - 65 | |
| Pulsed Drain Current | | $I_{DM}$ | - 240 | |
| Avalanche Current | | $I_{AR}$ | - 60 | |
| Repetitive Avalanche Energy[b] | L = 0.1 mH | $E_{AR}$ | 180 | mJ |
| Power Dissipation | $T_C$ = 25 °C (TO-220AB and TO-263) | $P_D$ | 187[d] | W |
| | $T_A$ = 25 °C (TO-263)[c] | | 3.75 | |
| Operating Junction and Storage Temperature Range | | $T_J$, $T_{stg}$ | - 55 to 175 | °C |

| THERMAL RESISTANCE RATINGS | | | | |
|---|---|---|---|---|
| **Parameter** | | **Symbol** | **Limit** | **Unit** |
| Junction-to-Ambient | PCB Mount (TO-263)[c] | $R_{thJA}$ | 40 | °C/W |
| | Free Air (TO-220AB) | | 62.5 | |
| Junction-to-Case | | $R_{thJC}$ | 0.8 | |

Notes:
a. Package limited.
b. Duty cycle ≤ 1 %.
c. When mounted on 1" square PCB (FR-4 material).
d. See SOA curve for voltage derating.

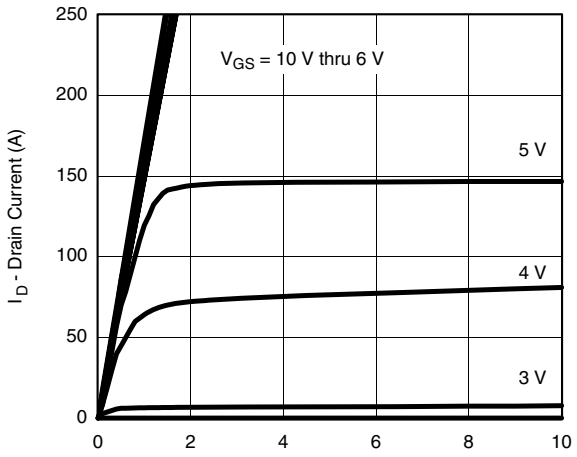
* Pb containing terminations are not RoHS compliant, exemptions may apply.

| **SPECIFICATIONS** ($T_J$ = 25 °C, unless otherwise noted) | | | | | | |
|---|---|---|---|---|---|---|
| **Parameter** | **Symbol** | **Test Conditions** | **Min.** | **Typ.** | **Max.** | **Unit** |
| **Static** | | | | | | |
| Drain-Source Breakdown Voltage | $V_{DS}$ | $V_{GS}$ = 0 V, $I_D$ = - 250 µA | - 30 | | | V |
| Gate Threshold Voltage | $V_{GS(th)}$ | $V_{DS}$ = $V_{GS}$, $I_D$ = - 250 µA | - 1 | | - 3 | |
| Gate-Body Leakage | $I_{GSS}$ | $V_{DS}$ = 0 V, $V_{GS}$ = ± 20 V | | | ± 100 | nA |
| Zero Gate Voltage Drain Current | $I_{DSS}$ | $V_{DS}$ = - 30 V, $V_{GS}$ = 0 V | | | - 1 | µA |
| | | $V_{DS}$ = - 30 V, $V_{GS}$ = 0 V, $T_J$ = 125 °C | | | - 50 | |
| | | $V_{DS}$ = - 30 V, $V_{GS}$ = 0 V, $T_J$ = 175 °C | | | - 250 | |
| On-State Drain Current[a] | $I_{D(on)}$ | $V_{DS}$ = - 5 V, $V_{GS}$ = - 10 V | - 120 | | | A |
| Drain-Source On-State Resistance[a] | $R_{DS(on)}$ | $V_{GS}$ = - 10 V, $I_D$ = - 30 A | | 0.0055 | 0.007 | Ω |
| | | $V_{GS}$ = - 10 V, $I_D$ = - 30 A, $T_J$ = 125 °C | | | 0.010 | |
| | | $V_{GS}$ = - 10 V, $I_D$ = - 30 A, $T_J$ = 175 °C | | | 0.013 | |
| | | $V_{GS}$ = - 4.5 V, $I_D$ = - 20 A | | 0.008 | 0.010 | |
| Forward Transconductance[a] | $g_{fs}$ | $V_{DS}$ = - 15 V, $I_D$ = - 75 A | 20 | | | S |
| **Dynamic[b]** | | | | | | |
| Input Capacitance | $C_{iss}$ | $V_{GS}$ = 0 V, $V_{DS}$ = - 25 V, f = 1 MHz | | 9000 | | pF |
| Output Capacitance | $C_{oss}$ | | | 1565 | | |
| Reversen Transfer Capacitance | $C_{rss}$ | | | 715 | | |
| Total Gate Charge[c] | $Q_g$ | $V_{DS}$ = - 15 V, $V_{GS}$ = - 10 V, $I_D$ = - 75 A | | 160 | 240 | nC |
| Gate-Source Charge[c] | $Q_{gs}$ | | | 32 | | |
| Gate-Drain Charge[c] | $Q_{gd}$ | | | 30 | | |
| Turn-On Delay Time[c] | $t_{d(on)}$ | $V_{DD}$ = - 15 V, $R_L$ = 0.2 Ω<br>$I_D \cong$ - 75 A, $V_{GEN}$ = - 10 V, $R_g$ = 2.5 Ω | | 25 | 40 | ns |
| Rise Time[c] | $t_r$ | | | 225 | 360 | |
| Turn-Off Delay Time[c] | $t_{d(off)}$ | | | 150 | 240 | |
| Fall Time[c] | $t_f$ | | | 210 | 340 | |
| **Source-Drain Diode Ratings and Characteristics[b]** ($T_C$ = 25 °C) | | | | | | |
| Continuous Current | $I_S$ | | | | - 75 | A |
| Pulsed Current | $I_{SM}$ | | | | - 240 | |
| Forward Voltage[a] | $V_{SD}$ | $I_F$ = - 75 A, $V_{GS}$ = 0 V | | - 1.2 | - 1.5 | V |
| Reverse Recovery Time | $t_{rr}$ | $I_F$ = - 75 A, dI/dt = 100 A/µs | | 55 | 100 | ns |
| Peak Reverse Recovery Current | $I_{RM(REC)}$ | | | 2.5 | 5 | A |
| Reverse Recovery Charge | $Q_{rr}$ | | | 0.07 | 0.25 | µC |

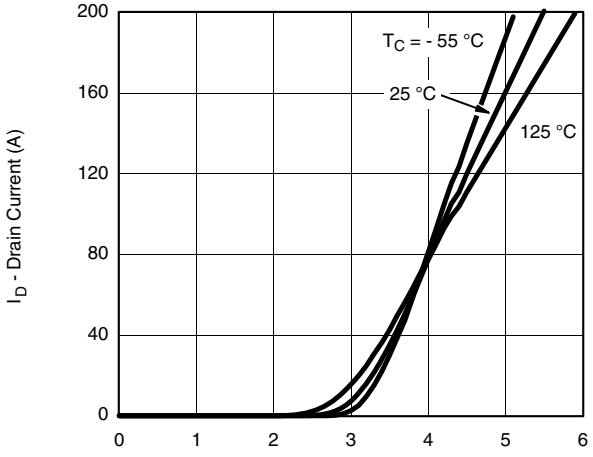
Notes:
a. Pulse test; pulse width ≤ 300 µs, duty cycle ≤ 2 %.
b. Guaranteed by design, not subject to production testing.
c. Independent of operating temperature.

*Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.*

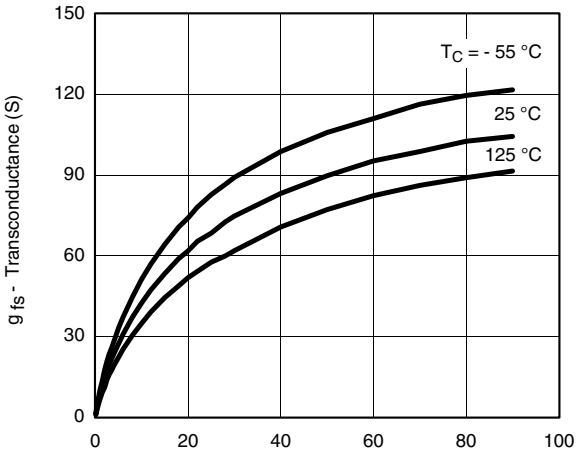
## TYPICAL CHARACTERISTICS (25 °C, unless otherwise noted)



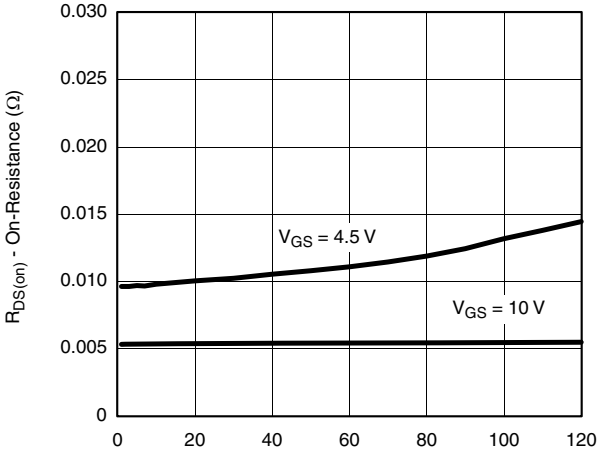

**Output Characteristics**



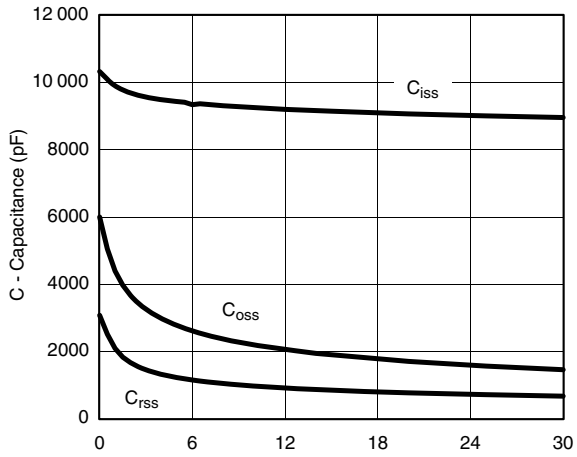

**Transfer Characteristics**



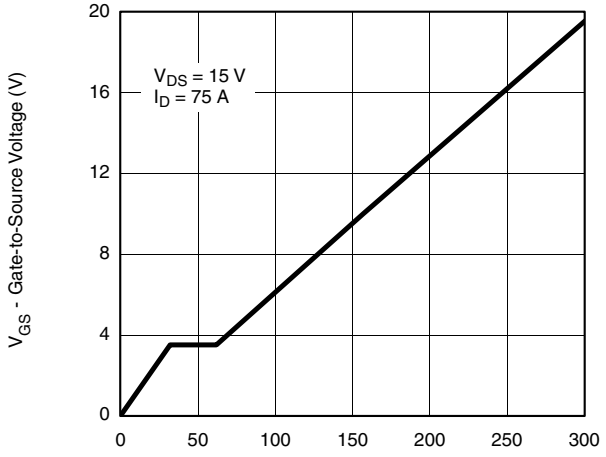

**Transconductance**




**On-Resistance vs. Drain Current**

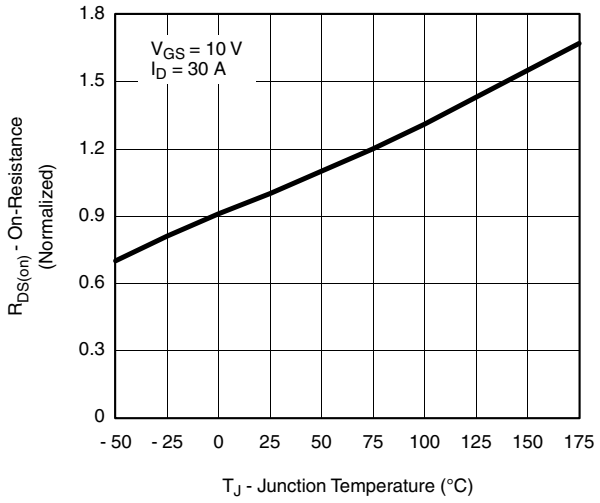



**Capacitance**

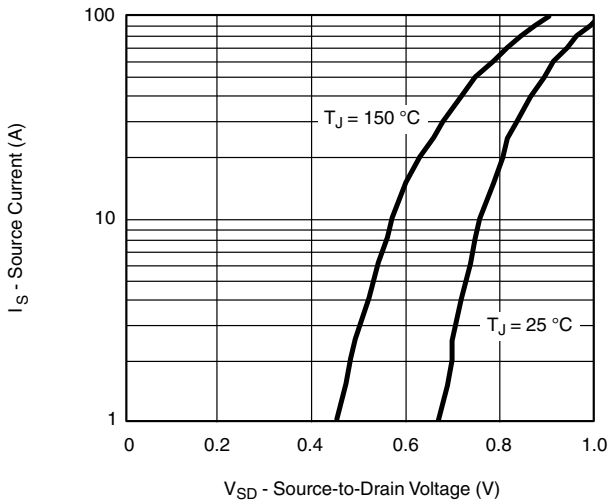



**Gate Charge**

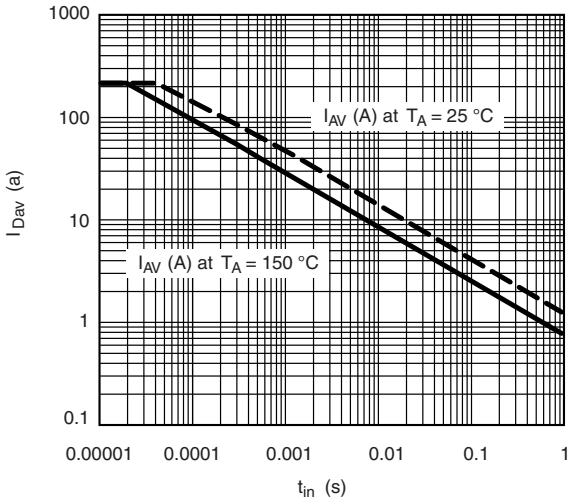
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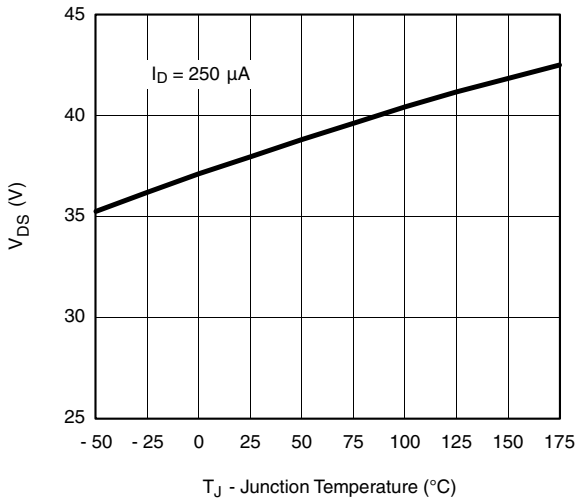

**On-Resistance vs. Junction Temperature**




**Source-Drain Diode Forward Voltage**

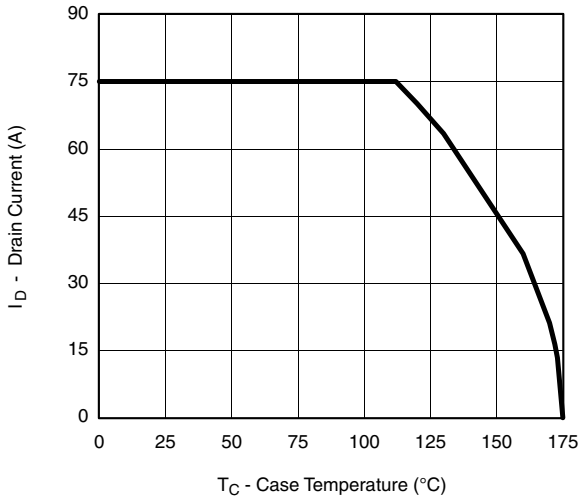



**Avalanche Current vs. Time**

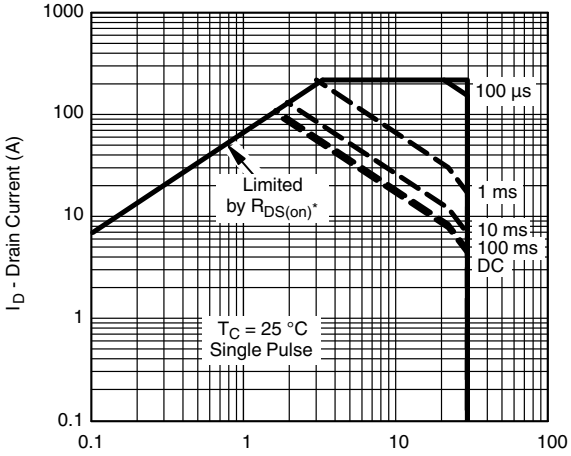



**Drain Source Breakdown vs. Junction Temperature**

## THERMAL RATINGS

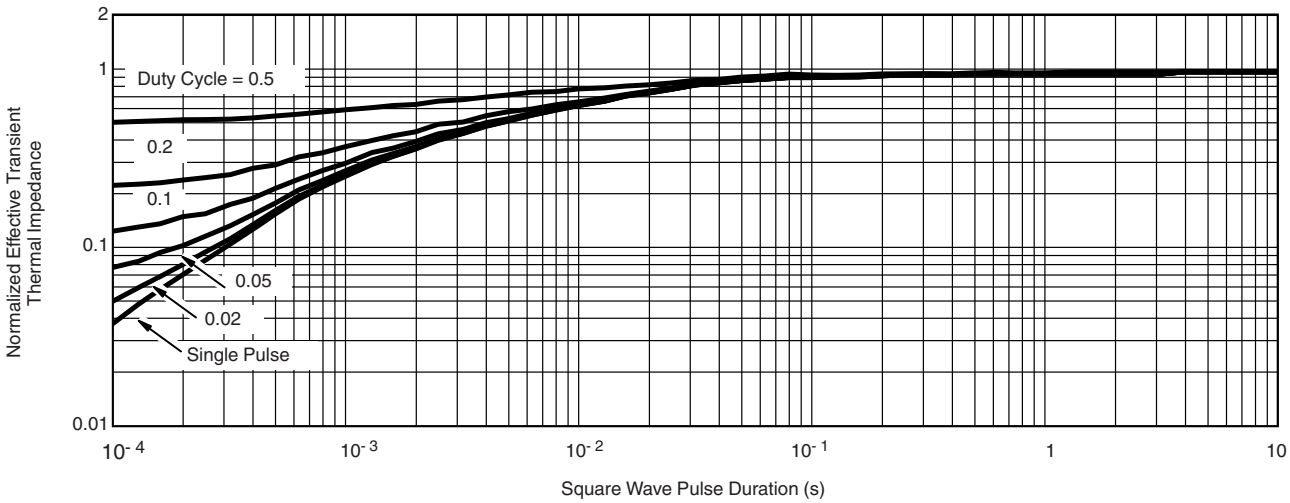



**Maximum Avalanche and Drain Current vs. Case Temperature**




* $V_{GS}$ > minimum $V_{GS}$ at which $R_{DS(on)}$ is specified

**Safe Operating Area**




**Normalized Thermal Transient Impedance, Junction-to-Case**

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